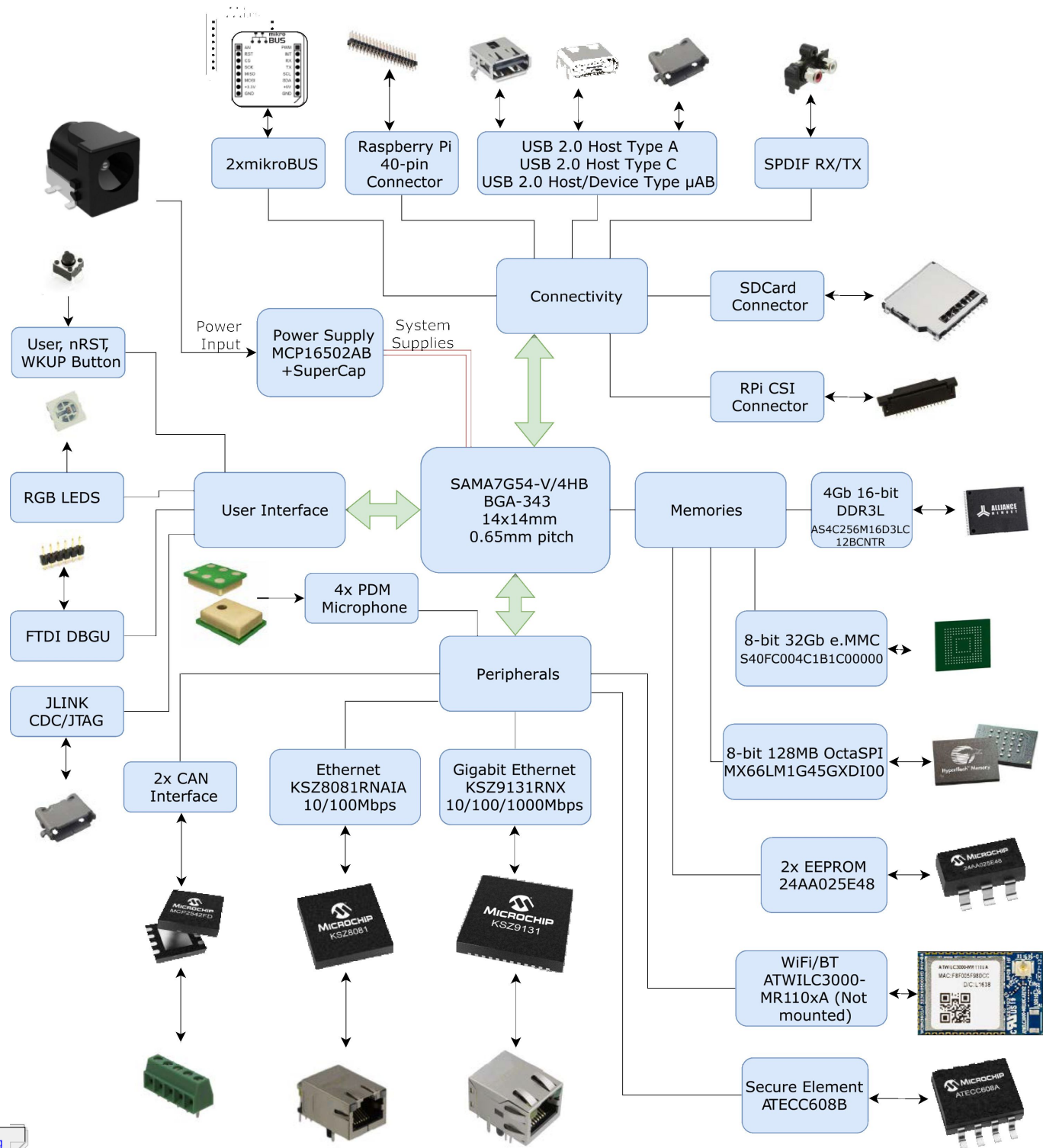


SAMA7G5-EK

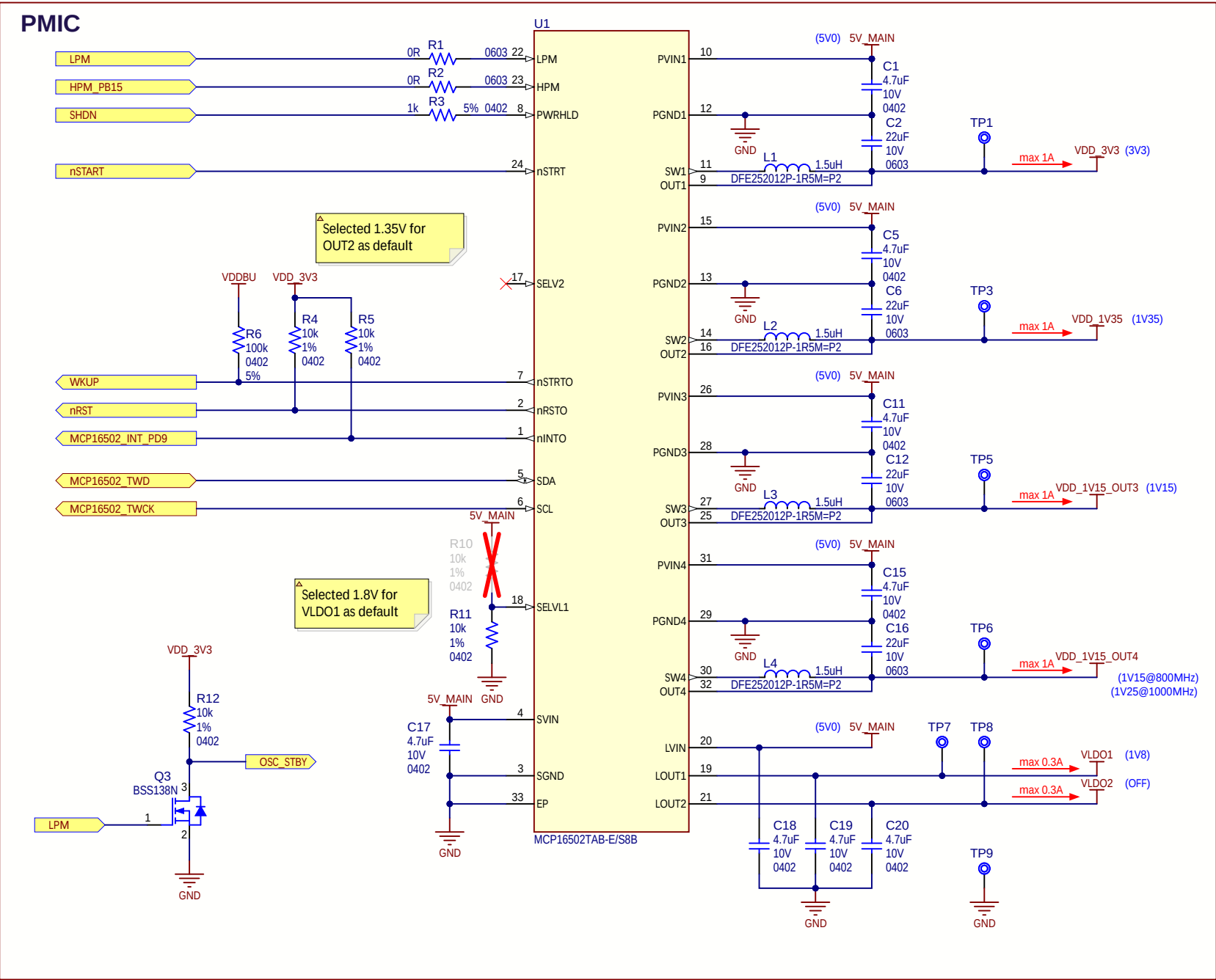
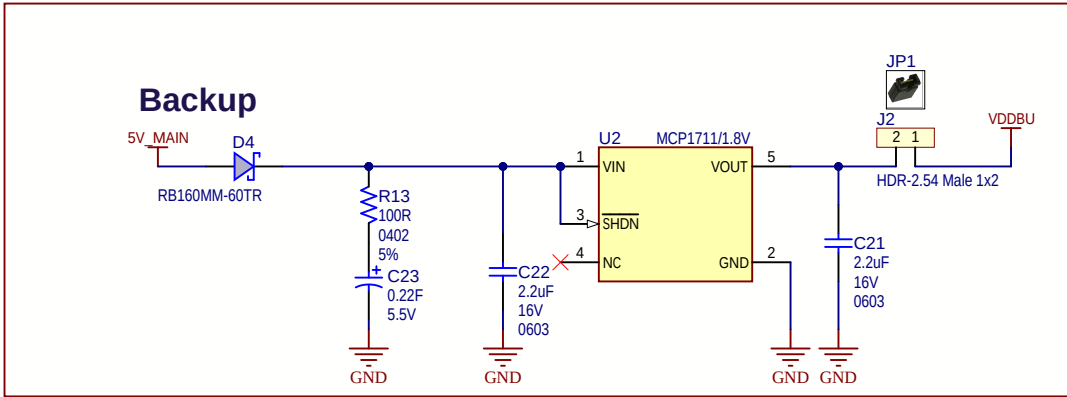
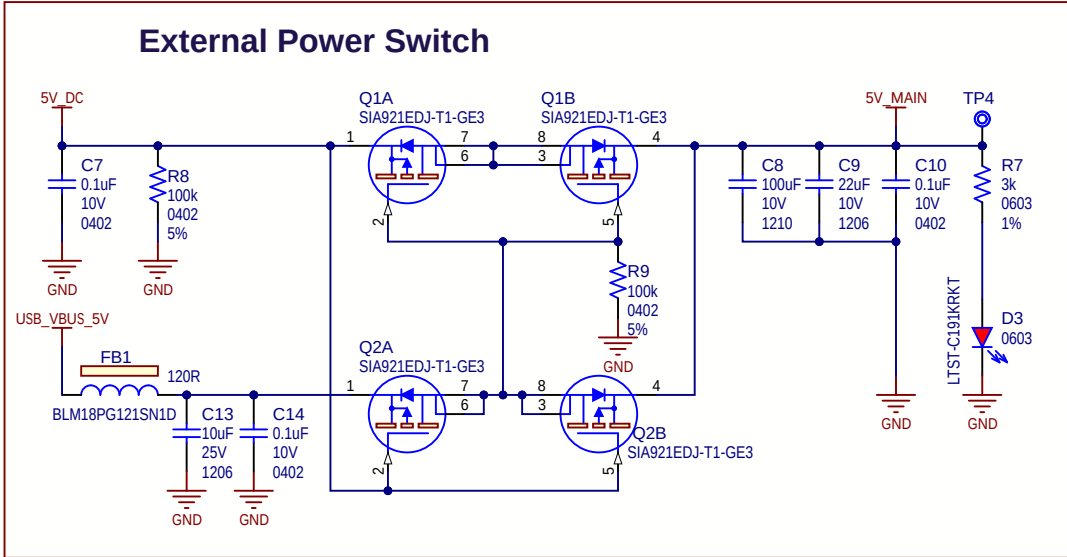
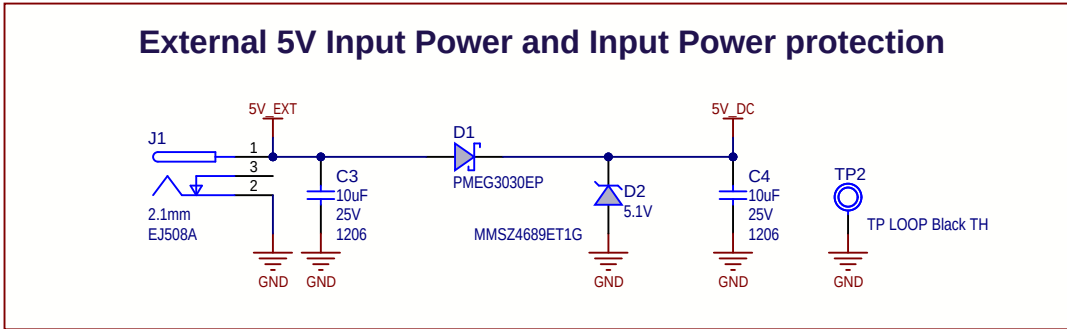


FIDUCIAL ROUND PCB 1mm SMD

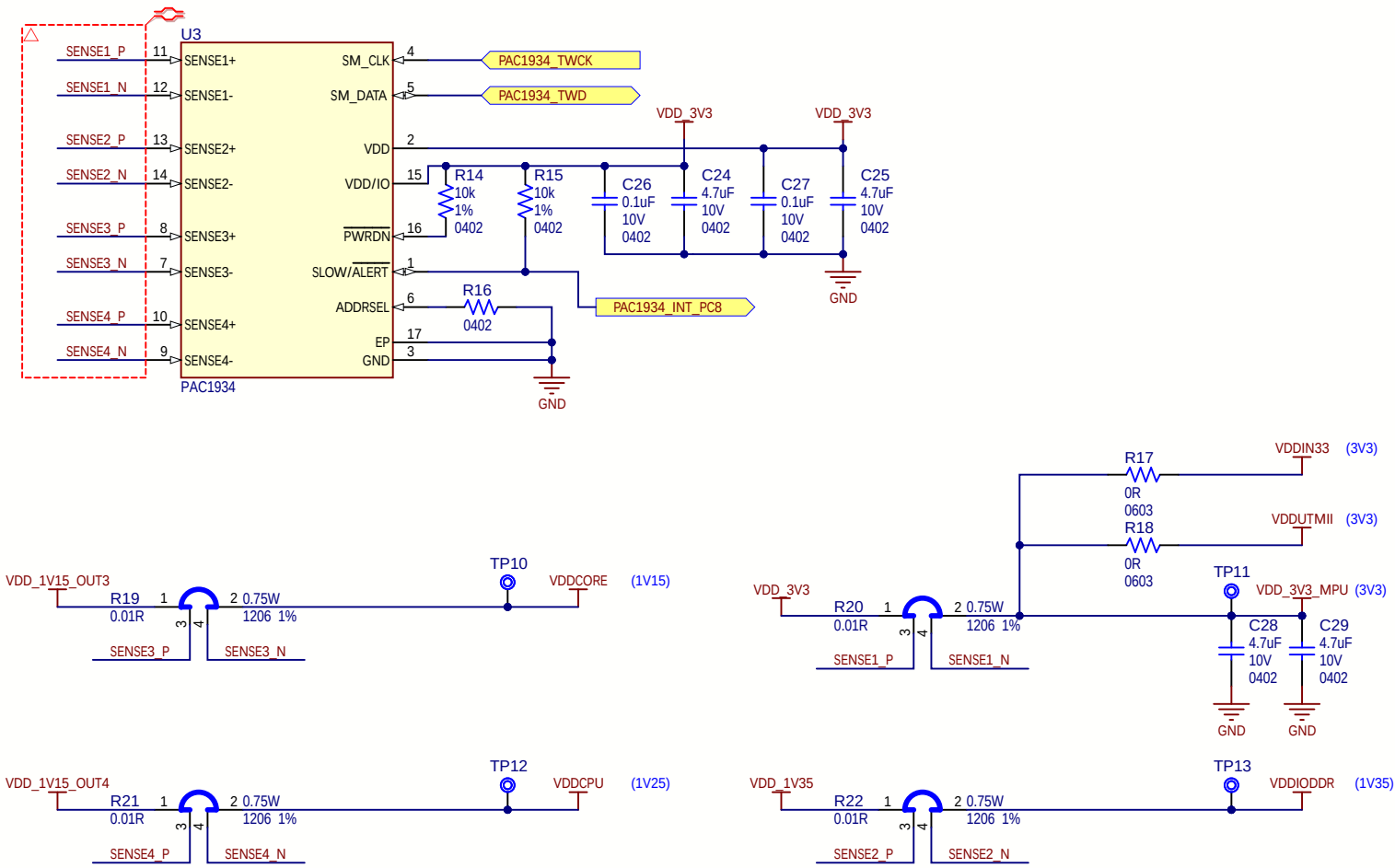
- FD1
- FD2
- FD3
- FD4

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Drawn By: CAV			
Engineer: CAV			
PartNumber: EV21H18A	Project Title SAMA7G54 Evaluation Kit		
Sheet Title Block Diagram	Variant Name Standard		Designed with 
Size B	SCH #: 03-00038	Rev: 5	Date: 3/1/2022 4:28:17 PM
	PCB #: 04-11152	Rev: 5	Sheet 1 of 15
File: 00 - SAMA7G54-Block Diagram.SchDoc			

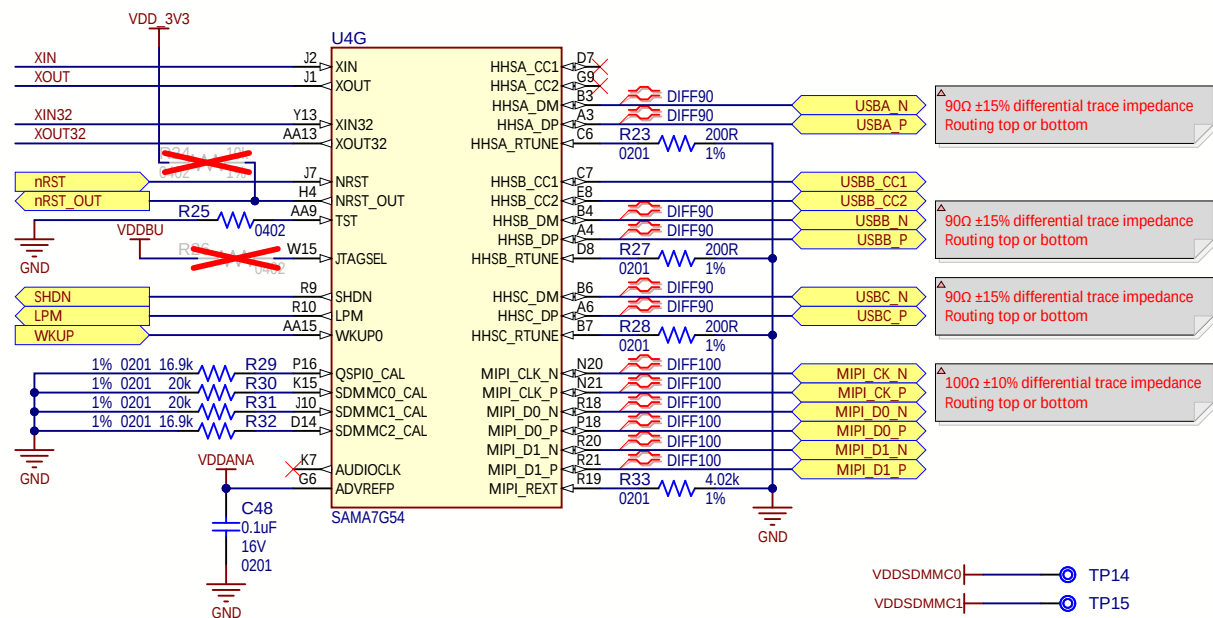


Power Measurement and Distribution

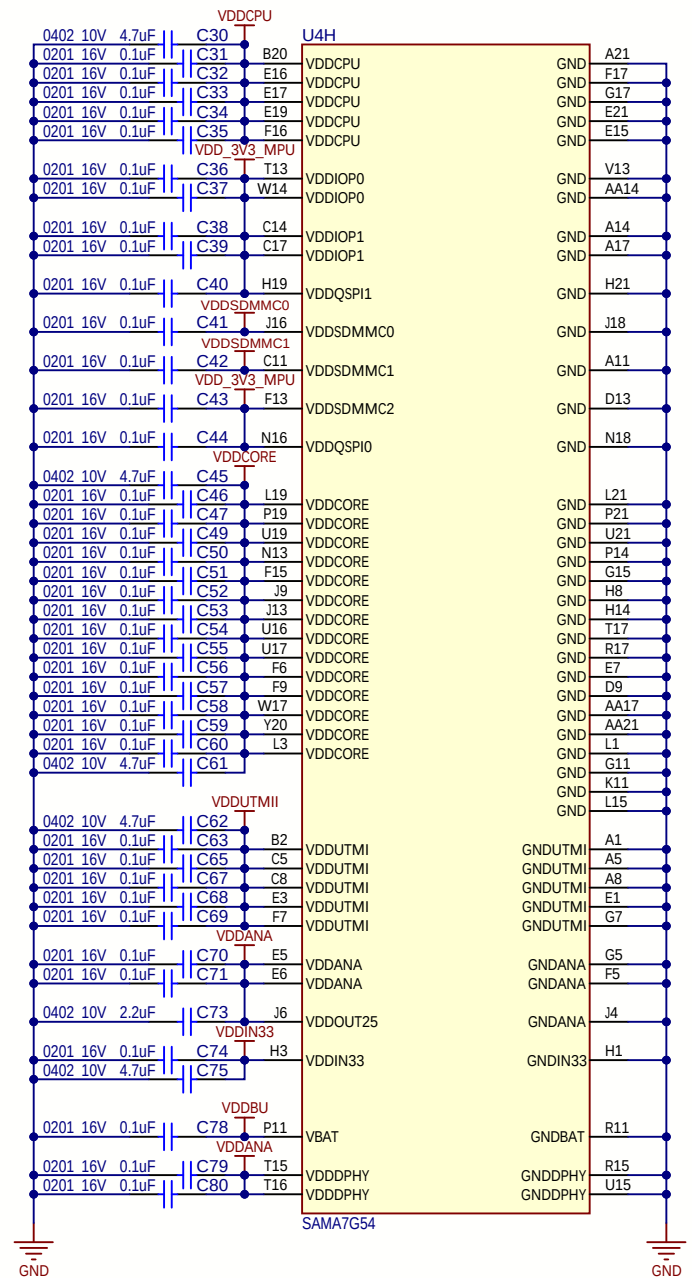


Drawn By: CAV			
Engineer: CAV			
PartNumber: EV21H18A	Project Title SAMA7G54 Evaluation Kit		
Sheet Title Power Distribution	Variant Name Standard	Designed with 	
Size B	SCH #: 03-00038	Rev: 5	Date: 3/1/2022 4:28:19 PM
	PCB #: 04-11152	Rev: 5	Sheet 3 of 15
File: 02 - Power Distribution.SchDoc			

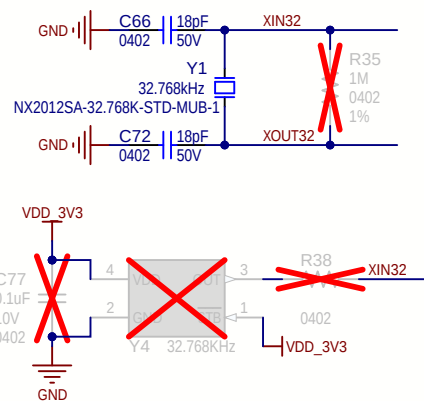
MPU configuration



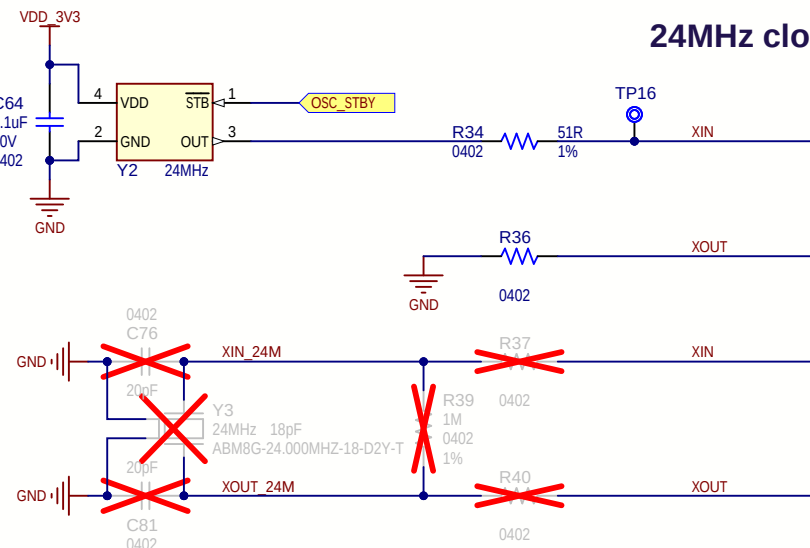
MPU decoupling



32.768KHz clock



24MHz clock

Drawn By:
CAVEngineer:
CAVPartNumber:
EV21H18ASheet Title
SAMA7G5-MPU

Size B

SCH #: 03-00038

PCB #: 04-11152

File: 03 - SAMA7G54-MPU.SchDoc

Project Title
SAMA7G54 Evaluation KitVariant Name
Standard

Date: 3/1/2022 4:28:19 PM

Rev: 5

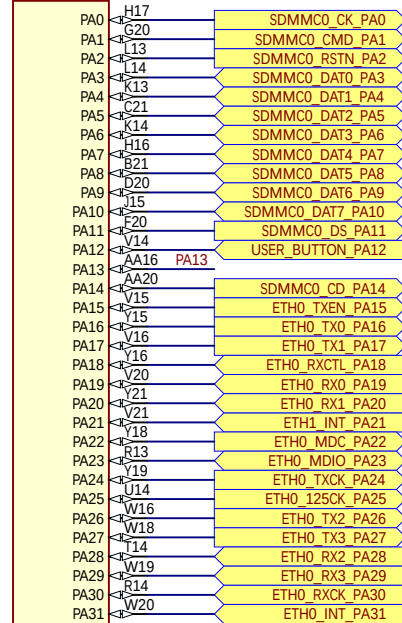
Sheet 4 of 15

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Altium

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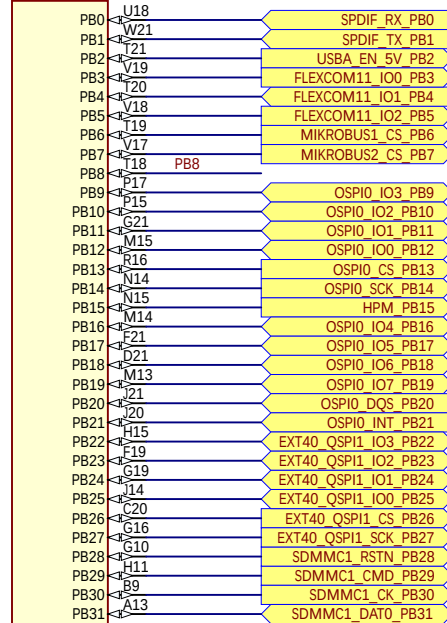
SAMA7G5 Bus Distribution

U4A



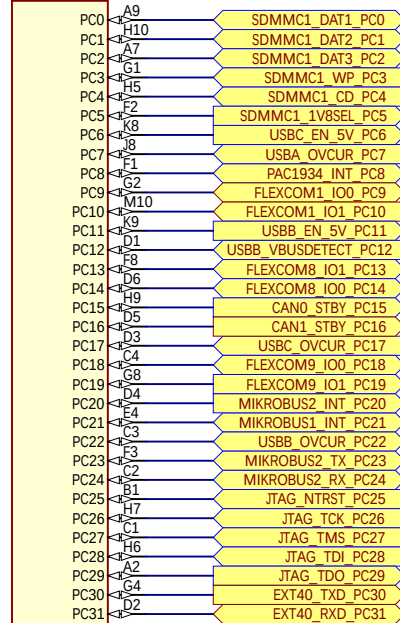
SAMA7G54

U4B



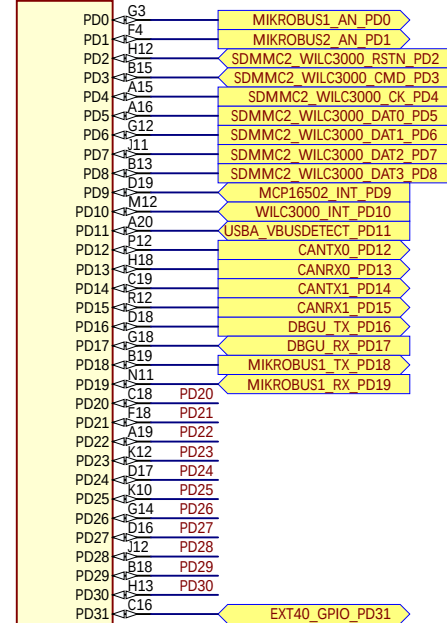
SAMA7G54

U4C



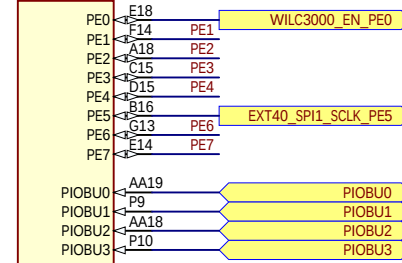
SAMA7G54

U4D

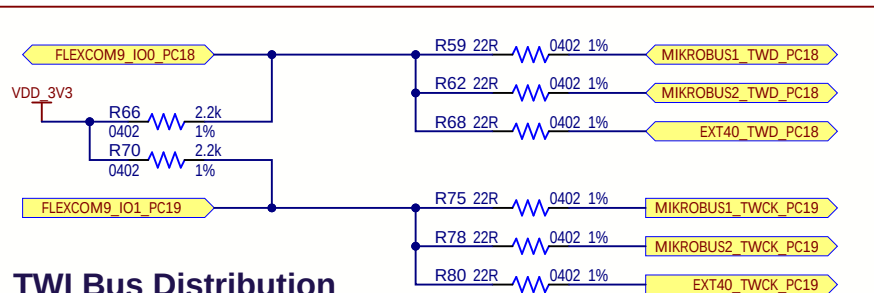


SAMA7G54

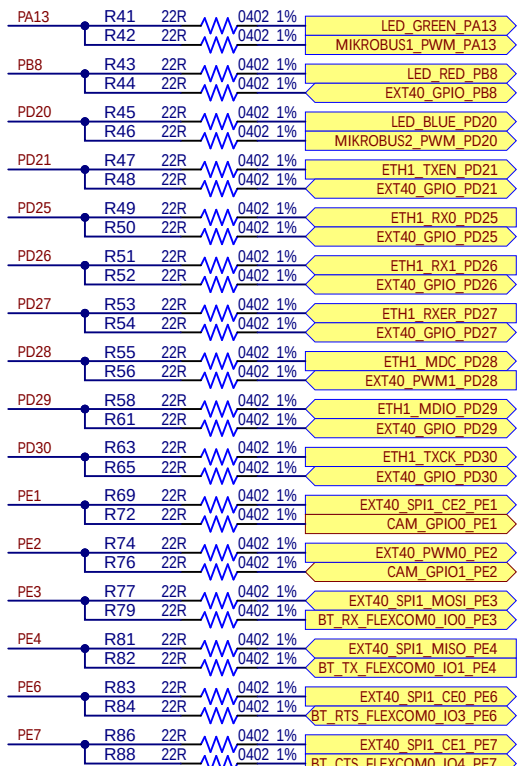
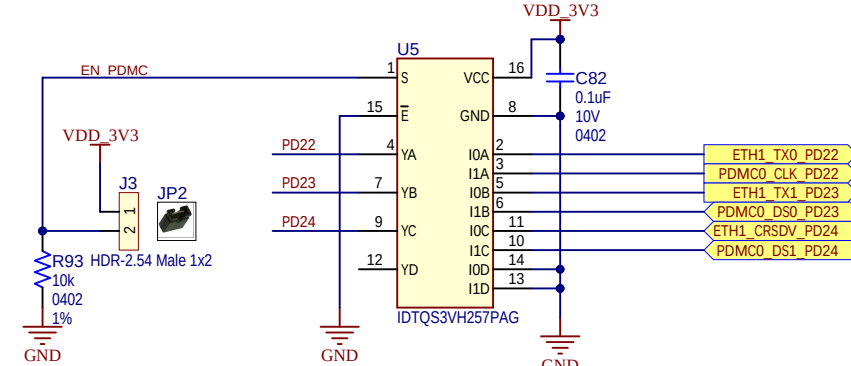
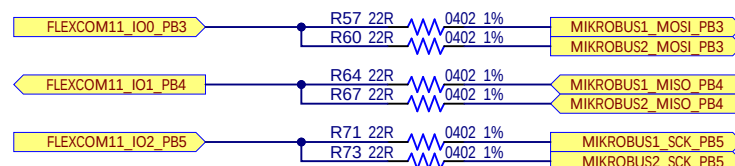
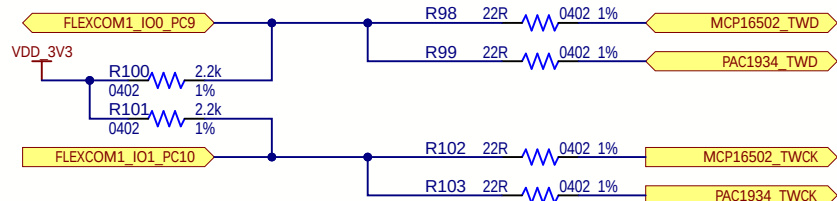
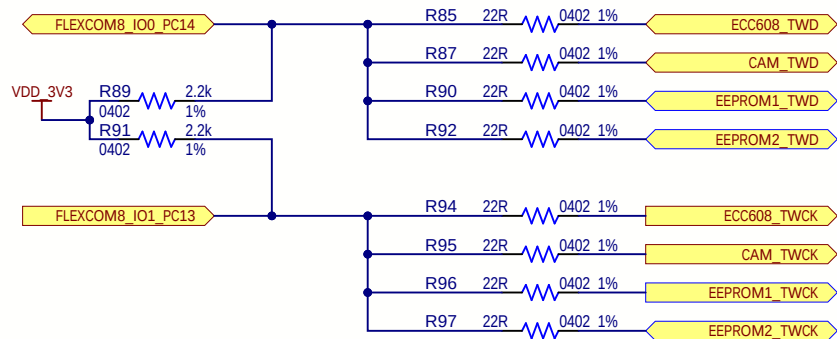
U4E



SAMA7G54



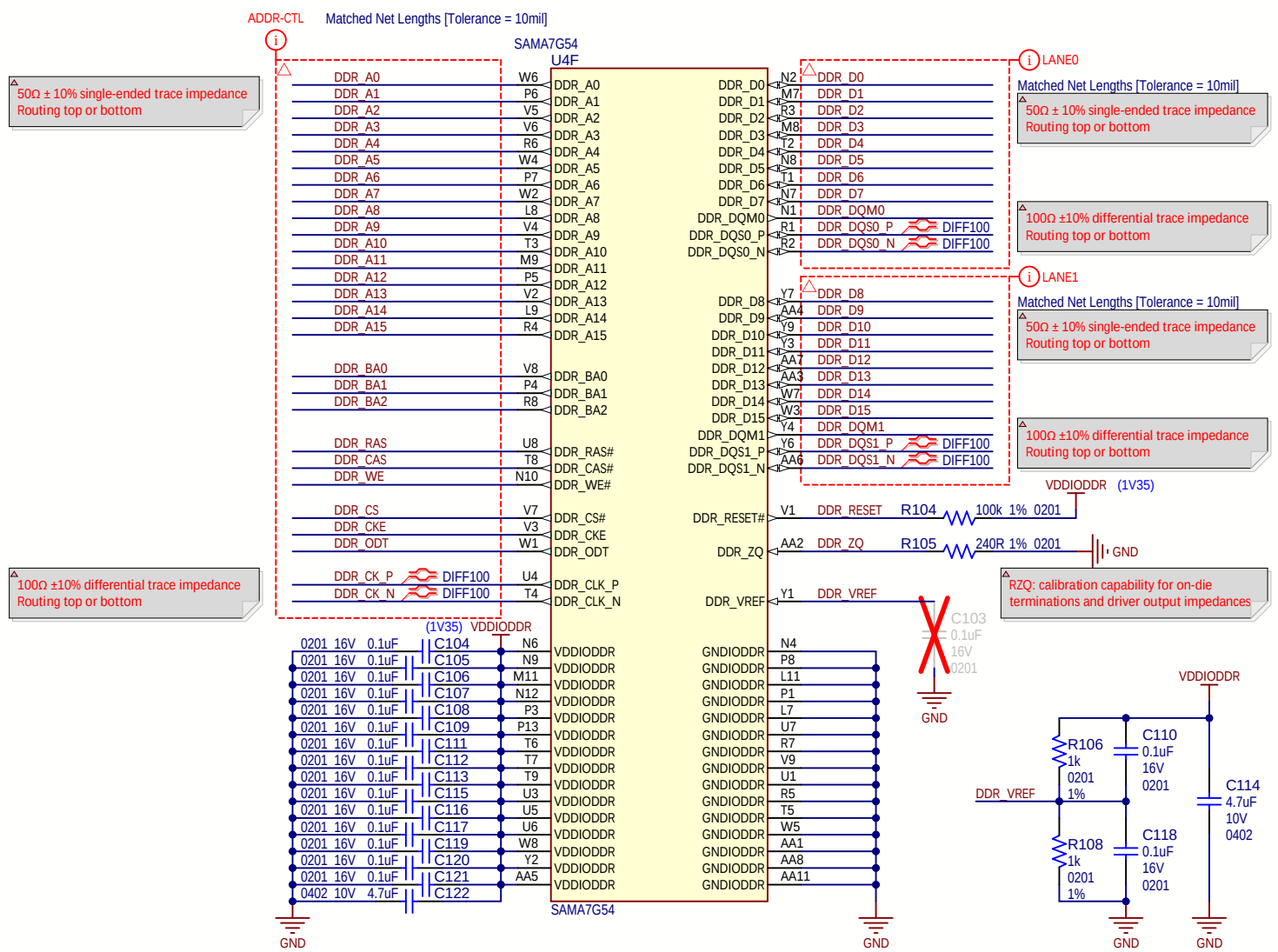
TWI Bus Distribution



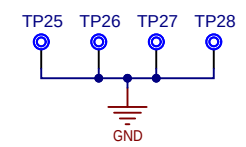
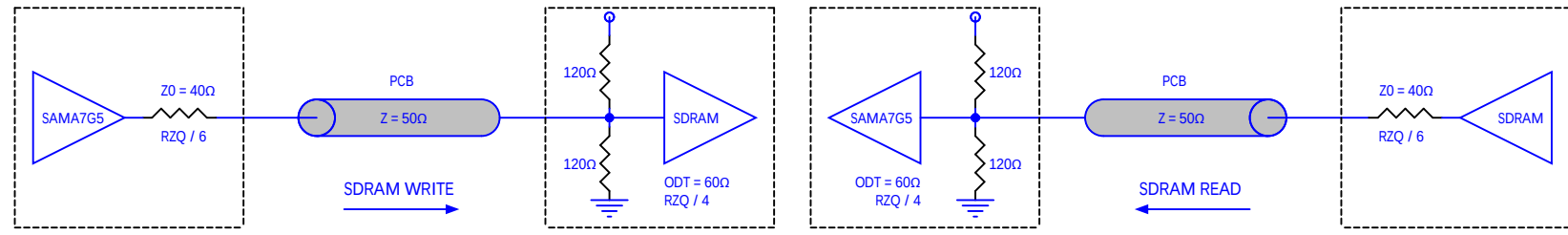
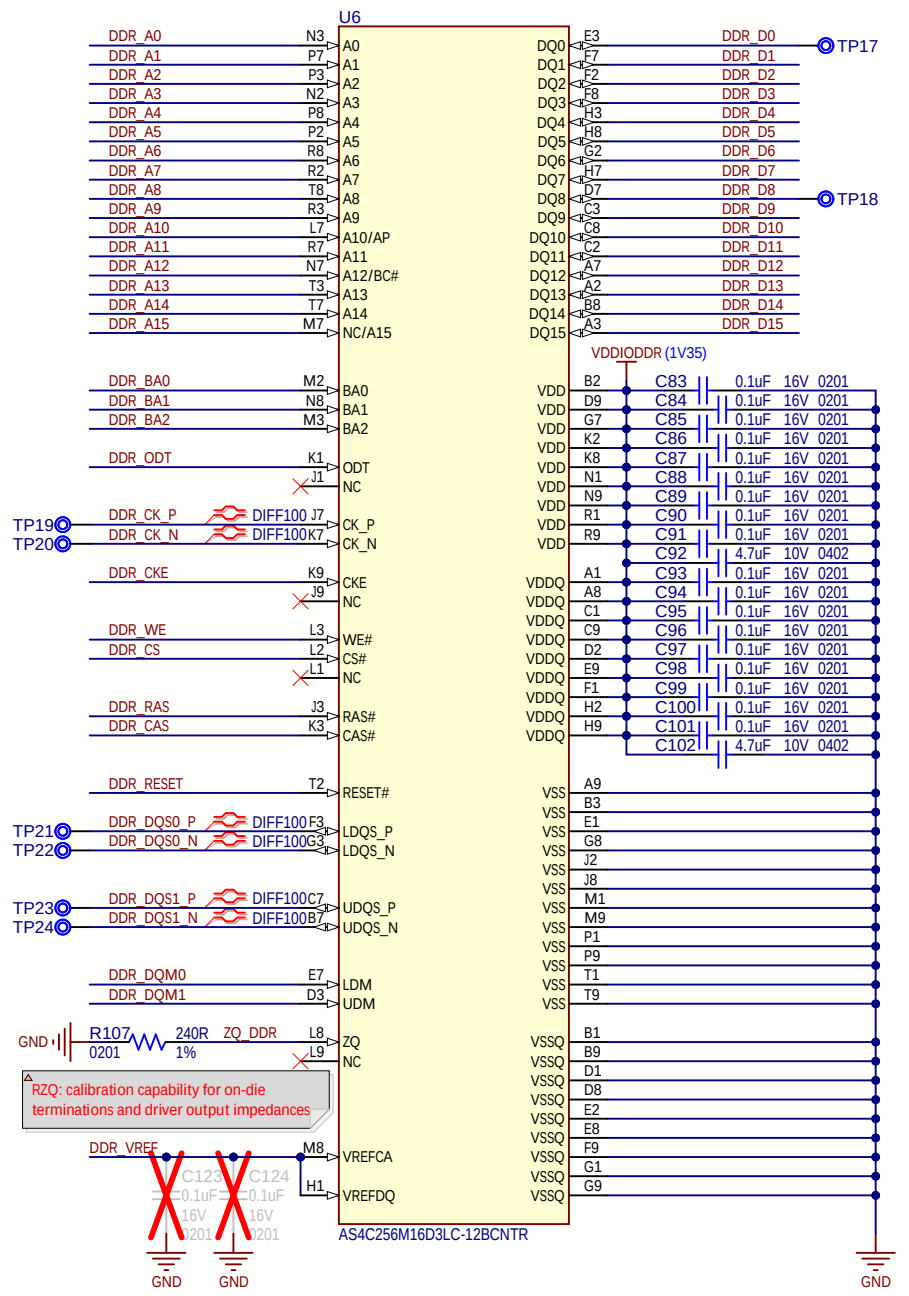
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Drawn By: CAV			
Engineer: CAV			
PartNumber: EV21H18A	Project Title SAMA7G54 Evaluation Kit		
Sheet Title SAMA7G5-PIO	Variant Name Standard	Designed with 	
Size B	SCH #: 03-00038	Rev: 5	Date: 3/1/2022 4:28:19 PM
	PCB #: 04-11152	Rev: 5	Sheet 5 of 15
File: 04 - SAMA7G54-PIO.SchDoc			

SAMA7G5 DDR3-1066 controller



4Gb DDR3L-1600 16bit



Drawn By:
CAV

Engineer:
CAV

PartNumber:
EV21H18A

Project Title
SAMA7G54 Evaluation Kit

Sheet Title
SAMA7G5-DDR3

Variant Name
Standard

Designed with
Altium

Size
B

SCH #: 03-00038

Rev: 5

Date: 3/1/2022 4:28:20 PM

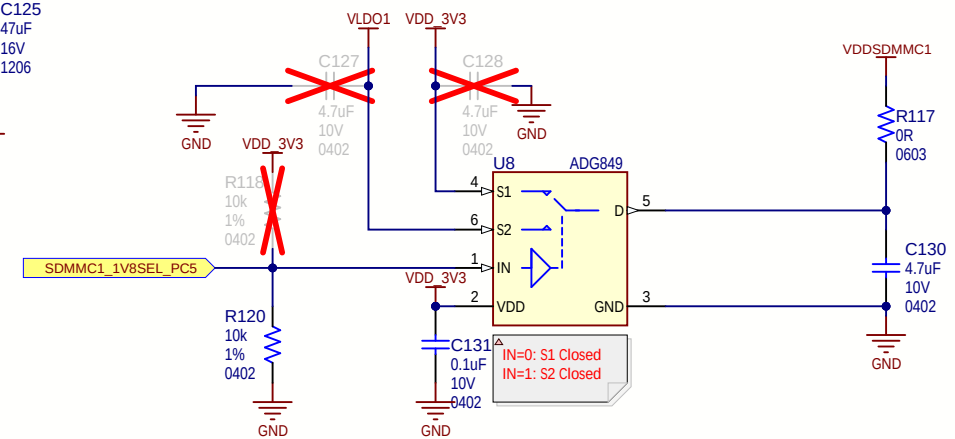
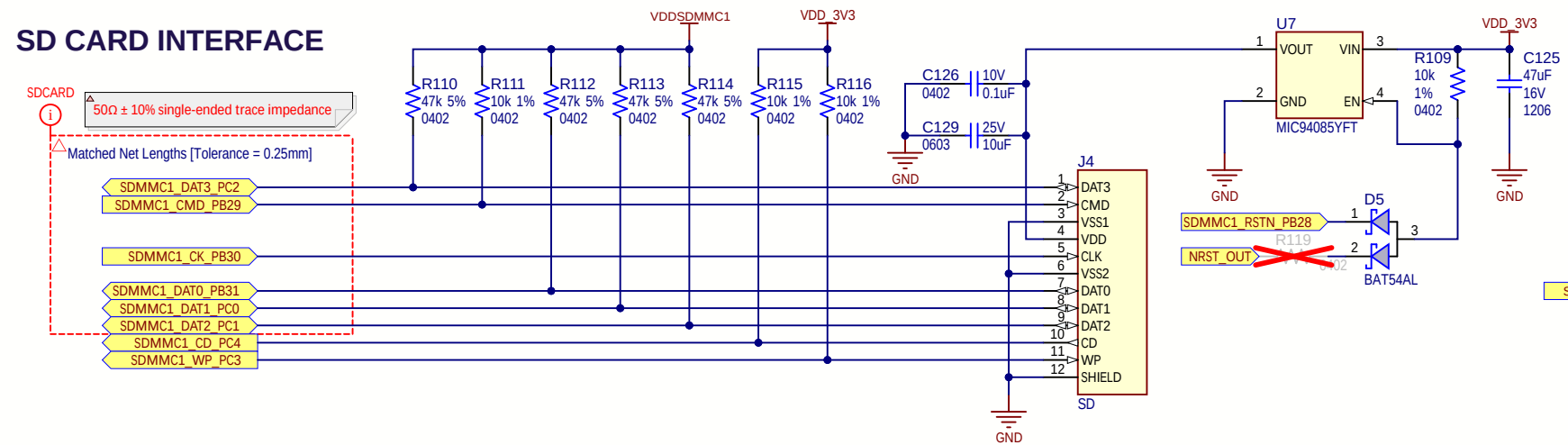
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Rev: 5

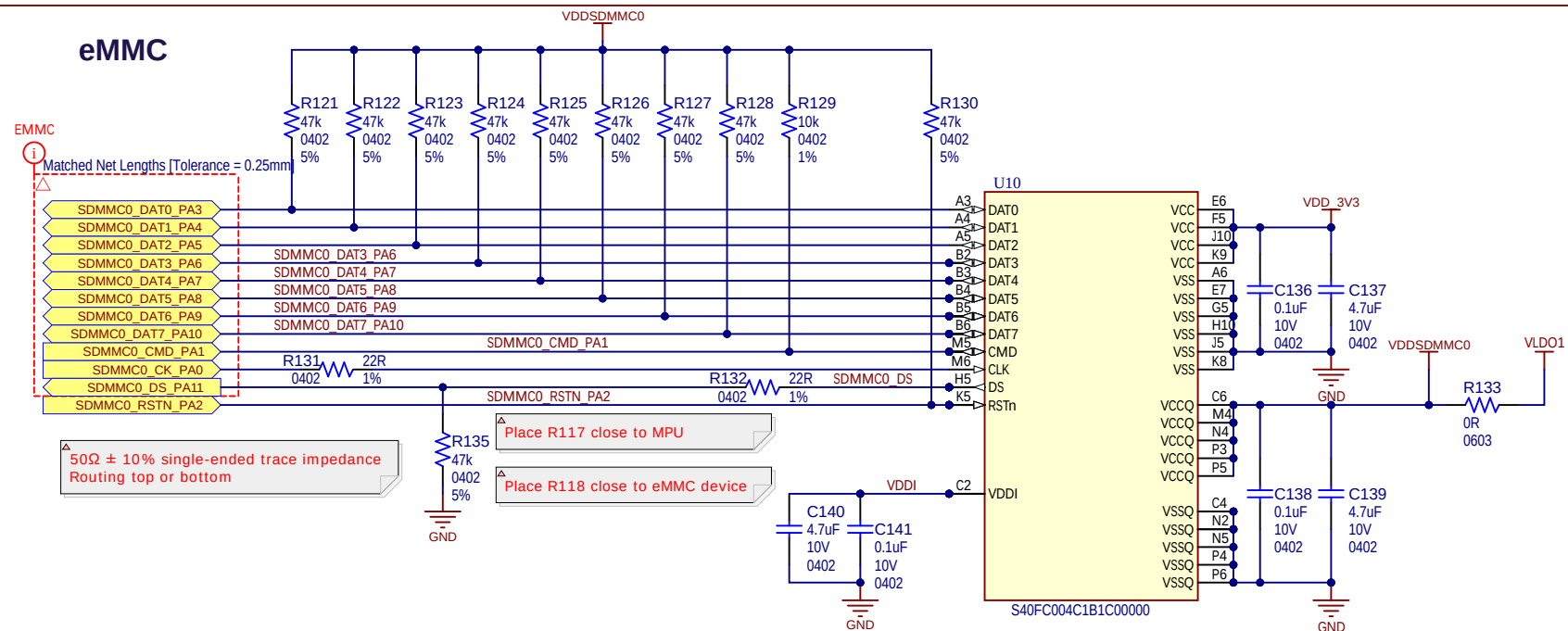
Sheet 6 of 15

File: 05 - SAMA7G54-DDR3.SchDoc

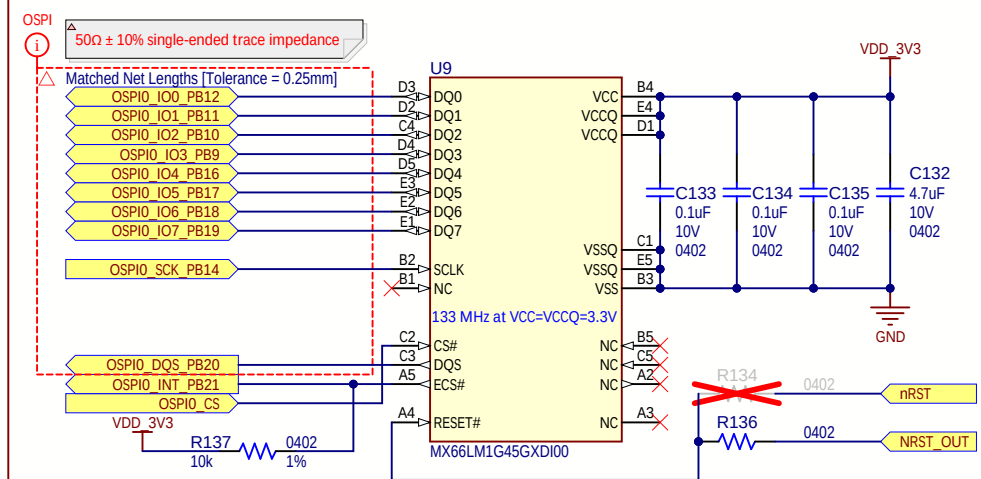
SD CARD INTERFACE



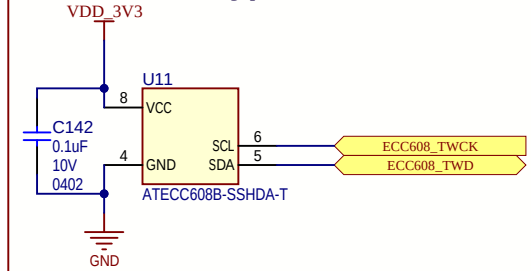
eMMC



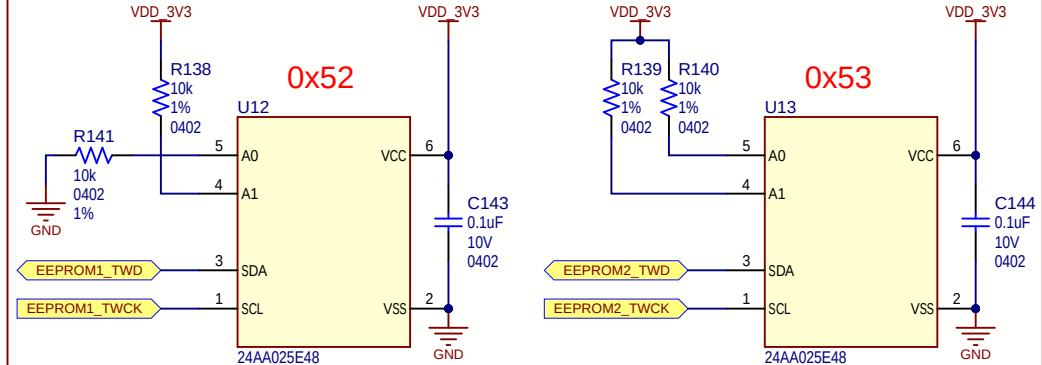
Octal SPI Memory



ECC608B CryptoAuthentication

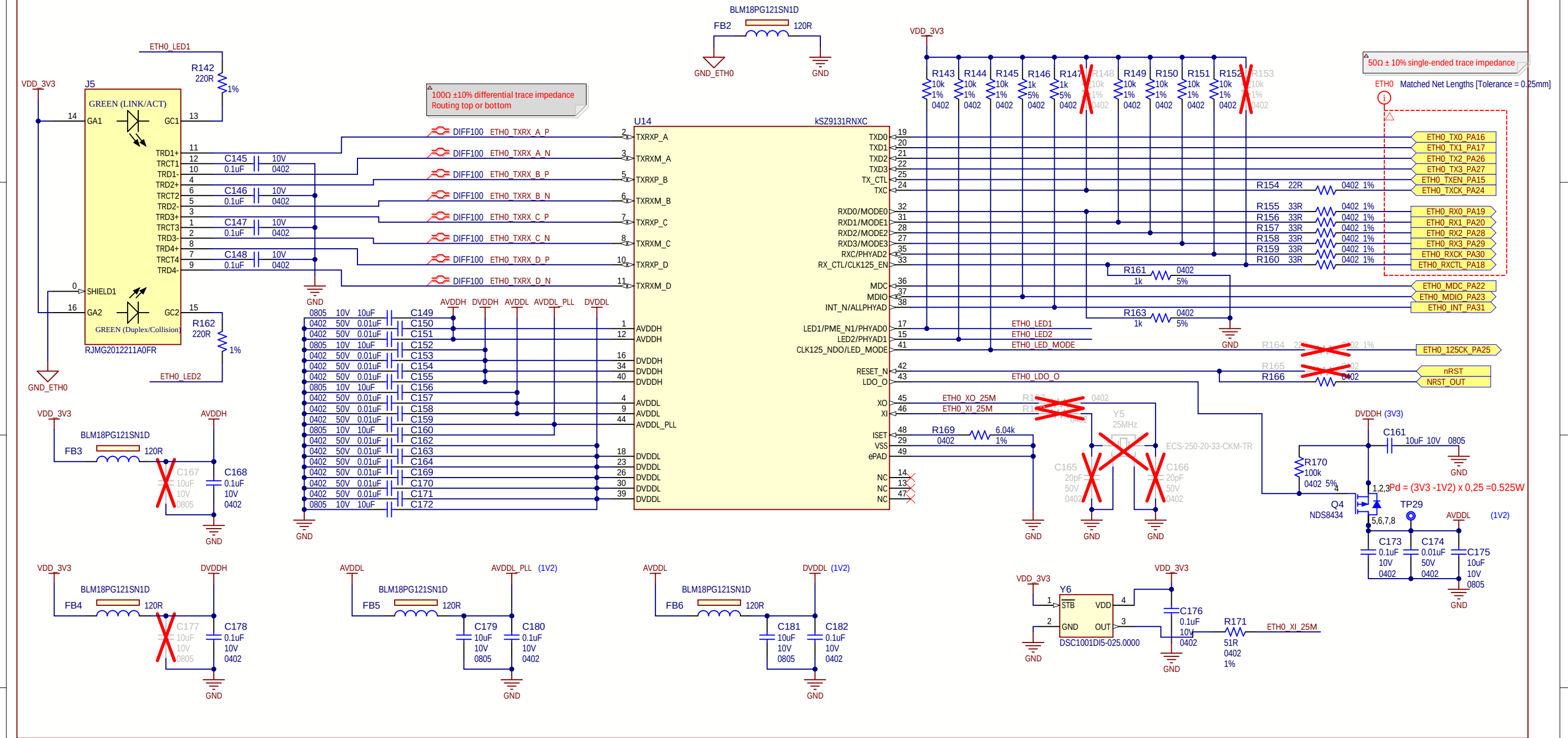


On Board EEPROM's



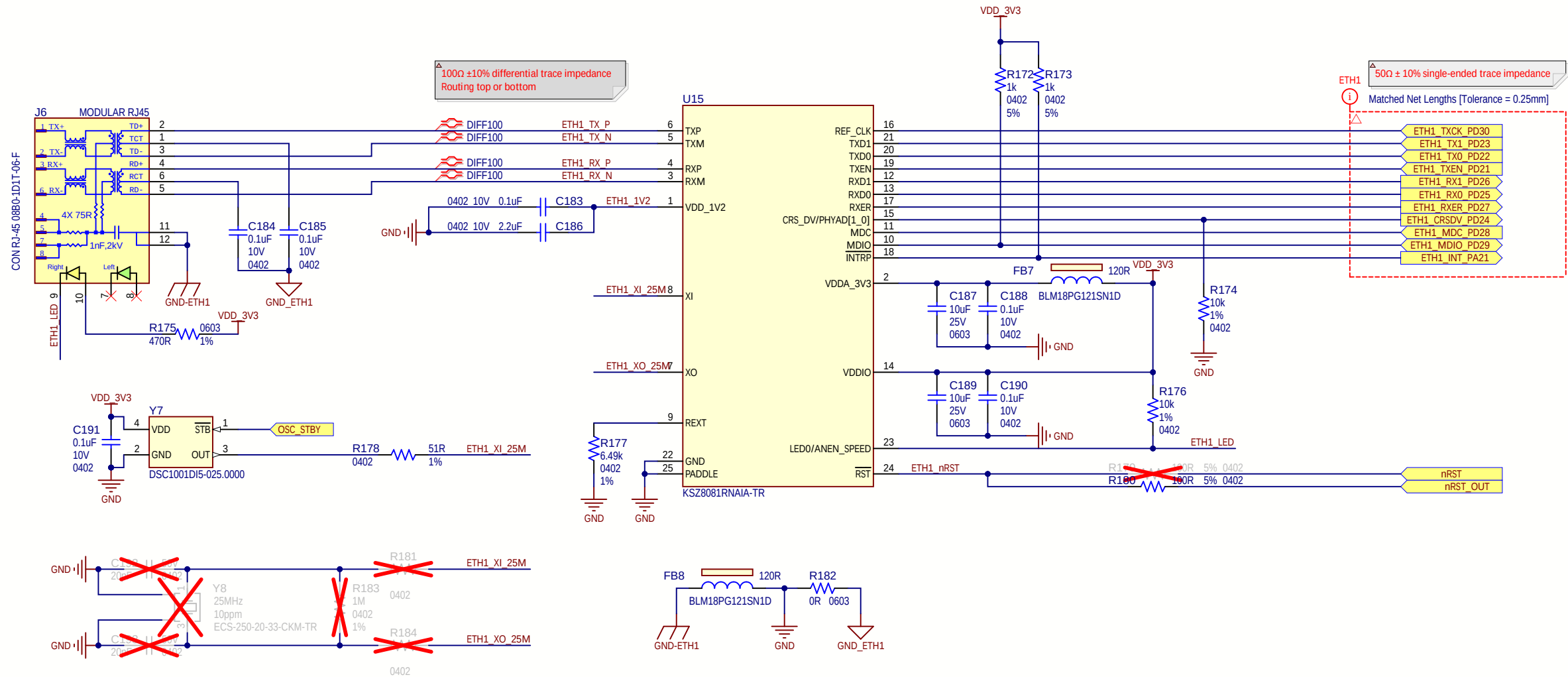
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Engineer: Cav			
PartNumber: EV21H18A		Project Title <i>SAMA7G54 Evaluation Kit</i>	
Sheet Title NVM Memories		Variant Name Standard	
Size B	SCH #: 03-00038	Rev: 5	Date: 3/1/2022 4:28:20 PM
	PCB #: 04-11152	Rev: 5	Sheet 7 of 15
File: 06 - NVM Memories.SchDoc		Designed with  Altium.com	

RGMII ETHERNET 1Gbps TRANSCEIVER



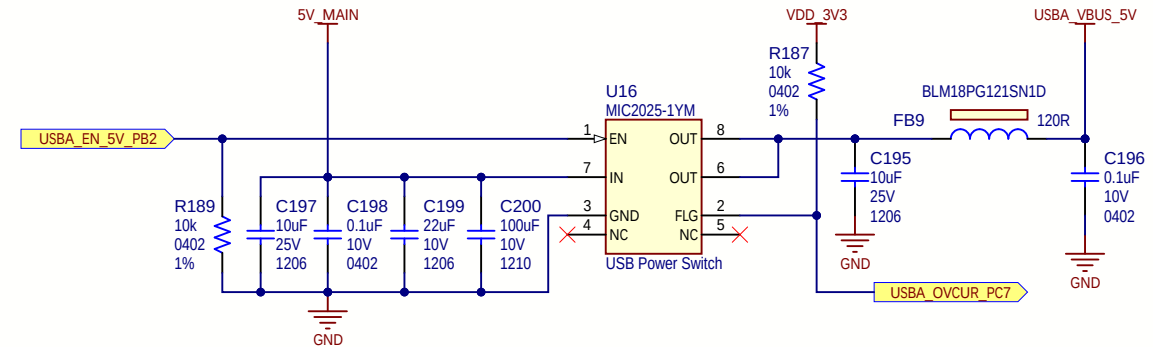
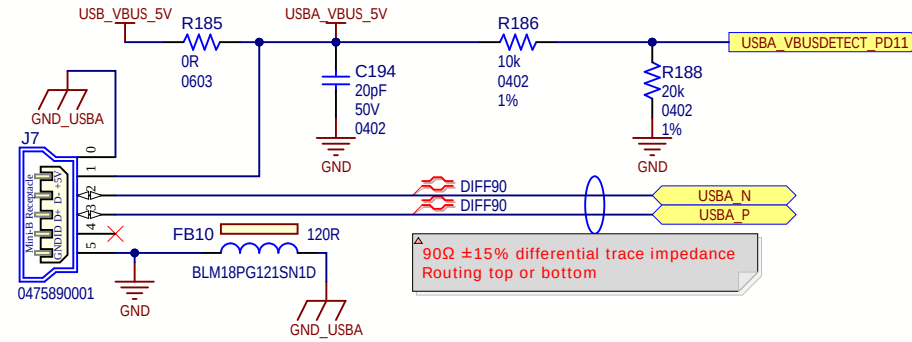
Drawn By: CAV	 MICROCHIP		
Engineer: CAV			
PartNumber: EV21H18A	Project Title SAMA7G54 Evaluation Kit		
Sheet Title ETH0-1000	Variant Name Standard		Designed with  Altium
Size B	SCH #: 03-00038	Rev: 5	Date: 3/1/2022 4:28:20 PM
	PCB #: 04-11152	Rev: 5	Sheet 8 of 15
File: 07 - ETH0-1000.SchDoc			

RMII ETHERNET 100Mbps TRANSCEIVER

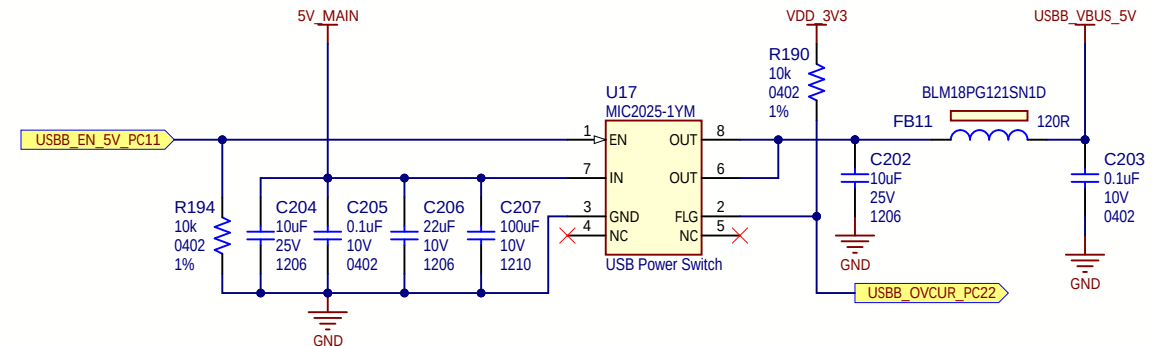
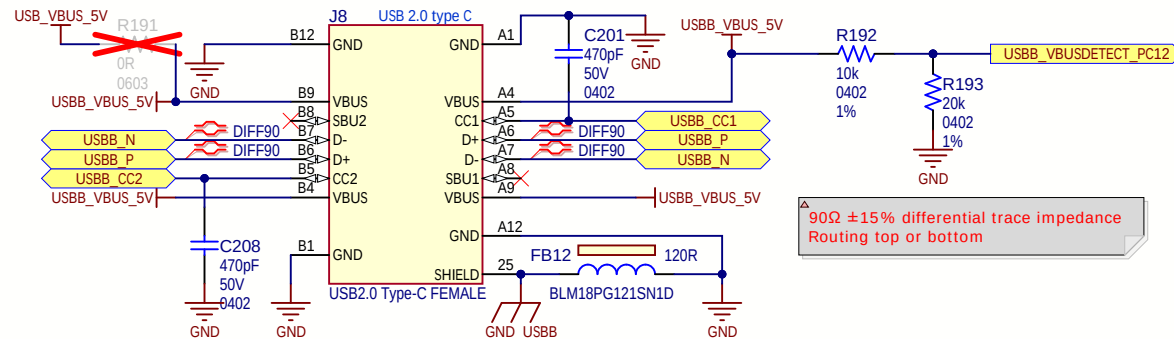


Drawn By: CAV	 MICROCHIP		
Engineer: CAV			
PartNumber: EV21H18A	Project Title SAMA7G54 Evaluation Kit		
Sheet Title ETH1-100	Variant Name Standard		Designed with  Altium
Size B	SCH #: 03-00038	Rev: 5	Date: 3/1/2022 4:28:20 PM
	PCB #: 04-11152	Rev: 5	Sheet 9 of 15
File: 08 - ETH1-100.SchDoc			

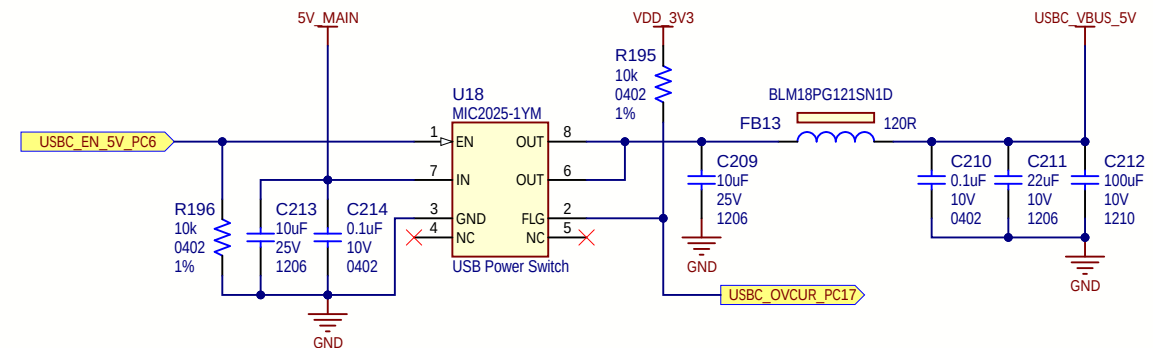
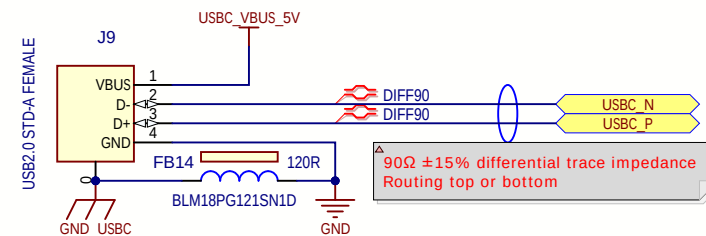
USB Micro-AB type connector to the USBA port (HOST/DEVICE)



USB-C type connector to the USBB port (HOST/DEVICE)



USB-A type connector the USBC port (HOST)



Drawn By:
CAV

Engineer:
CAV

PartNumber:
EV21H18A

Sheet Title
USB

Size	B
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File: 09 - USB.SchDoc



Project Title	<i>SAMA7G54 Evaluation Kit</i>
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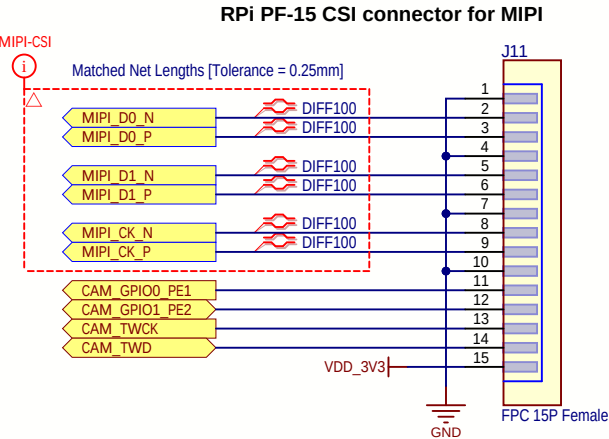
Variant Name	Standard
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Date: 3/1/2022 4:28:20 PM
Sheet 10 of 15

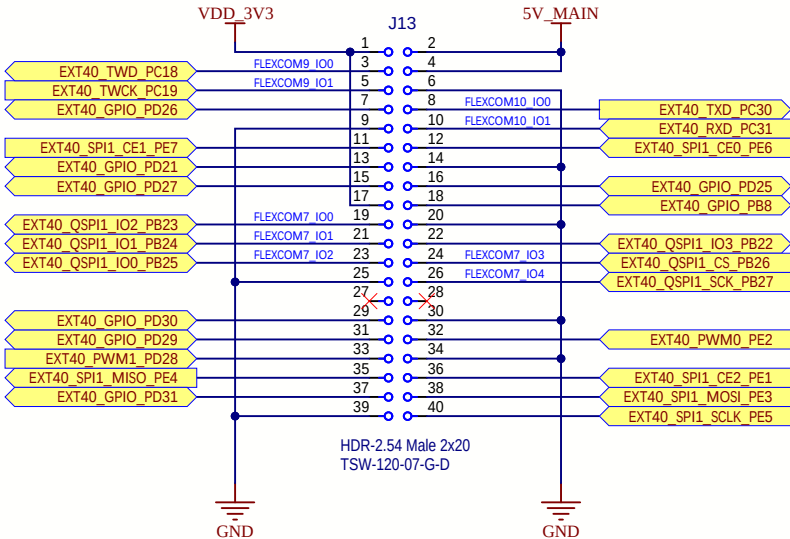
Designed with

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MIPI connector

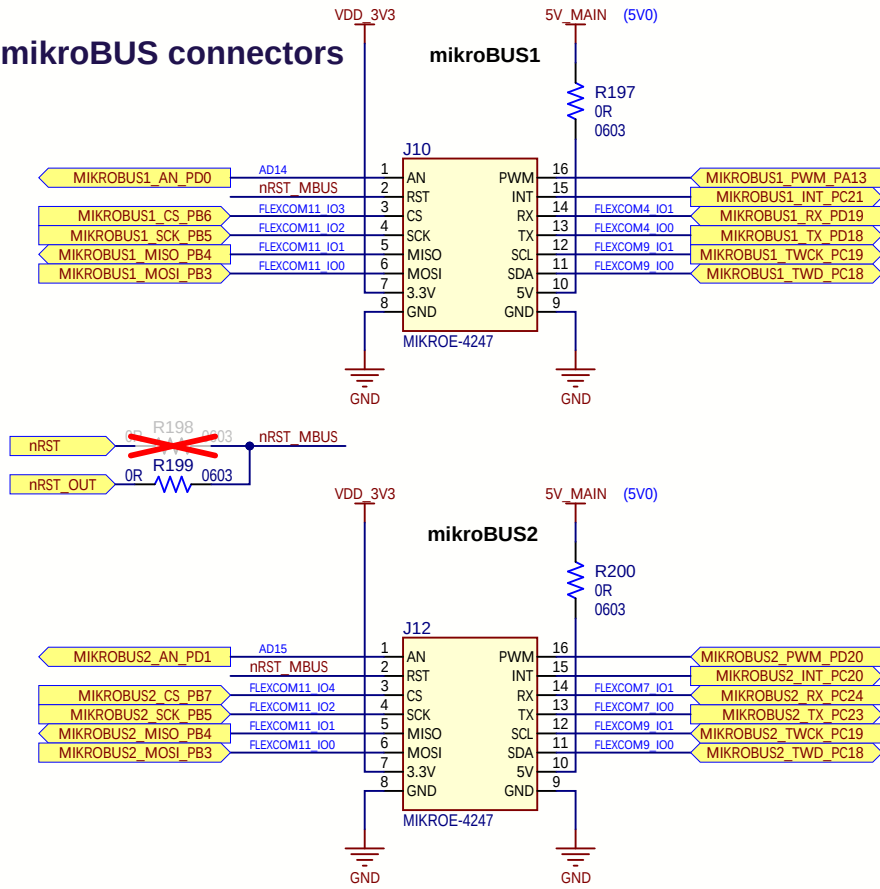


Raspberry Pi 40-pin Connector

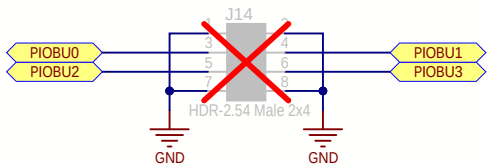


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mikroBUS connectors

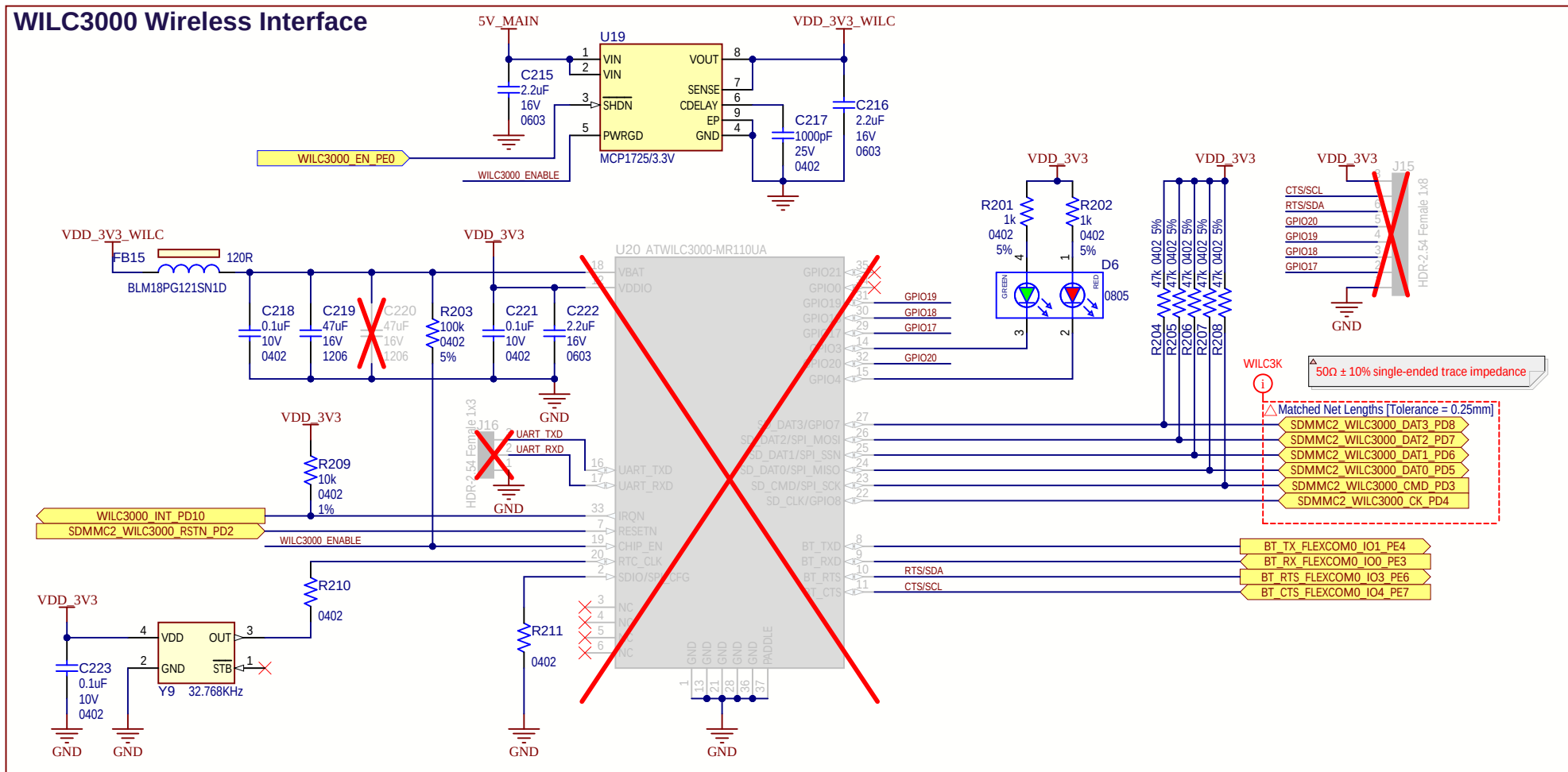


User available Tamper Pins

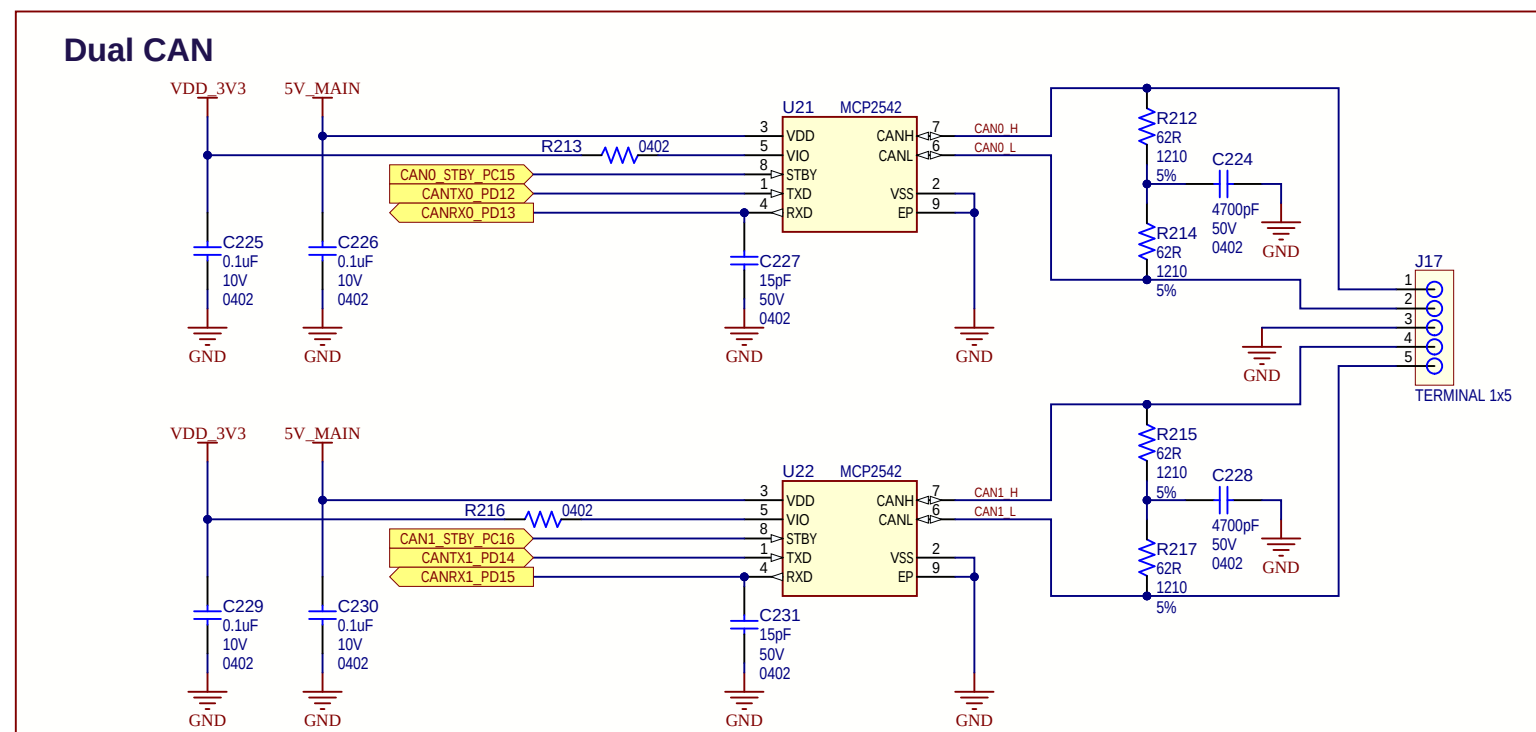


Drawn By: CAV	 MICROCHIP		
Engineer: CAV			
PartNumber: EV21H18A	Project Title SAMA7G54 Evaluation Kit		
Sheet Title Expansion Connectors	Variant Name Standard	Designed with  Altium	
Size B	SCH #: 03-00038	Rev: 5	Date: 3/1/2022 4:28:21 PM
	PCB #: 04-11152	Rev: 5	Sheet 11 of 15
File: 10 - Expansion Connectors.SchDoc			

WILC3000 Wireless Interface

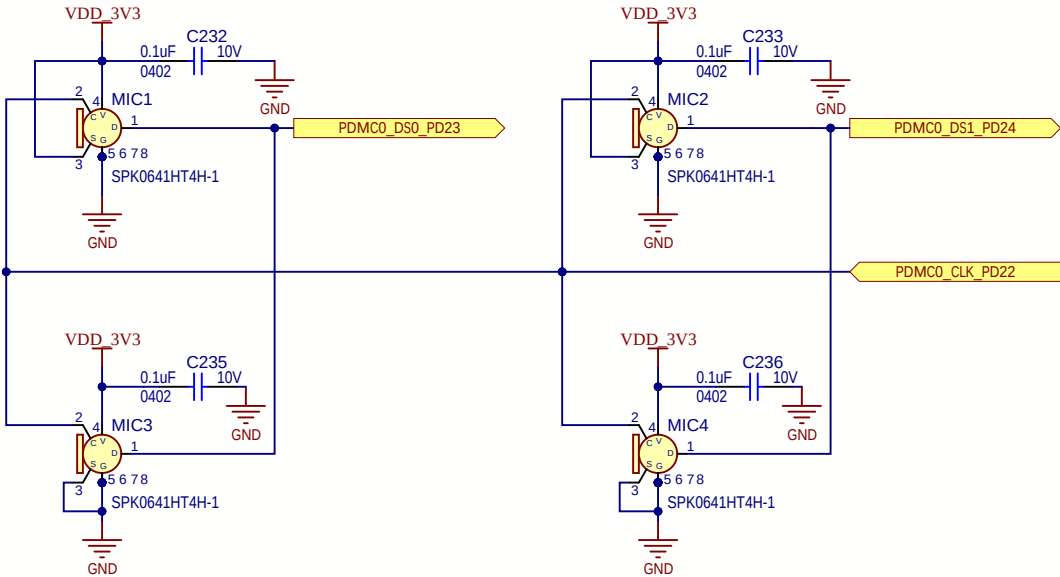


Dual CAN

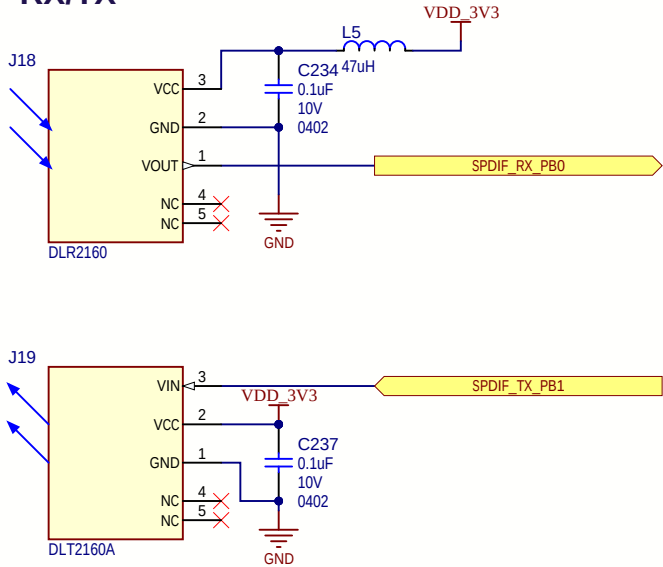


Drawn By: CAV		 MICROCHIP	
Engineer: CAV			
PartNumber: EV21H18A		Project Title <i>SAMA7G54 Evaluation Kit</i>	
Sheet Title Wireless - CAN		Variant Name Standard	
Size B	SCH #: 03-00038	Rev: 5	Date: 3/1/2022 4:28:21 PM
	PCB #: 04-11152	Rev: 5	Sheet 12 of 15
File: 11 - Wireless - CAN.SchDoc		Designed with  Altium.com	

4x PDM Microphones

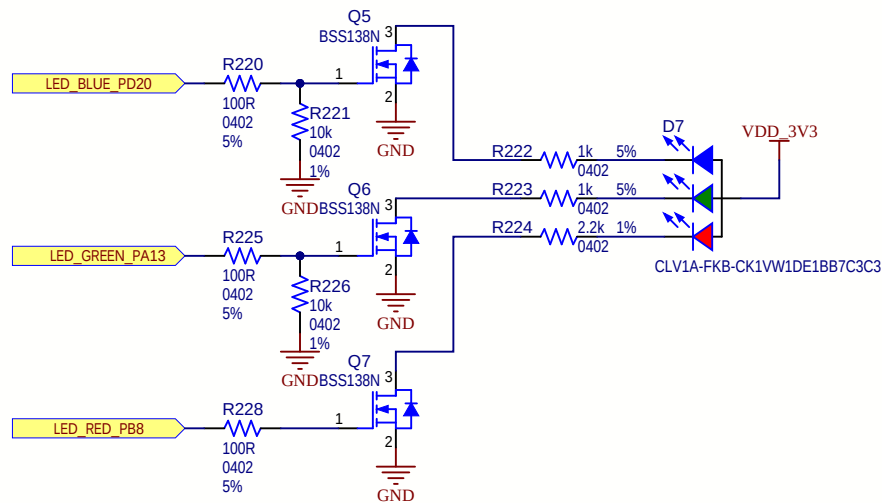


SPDIF RX/TX

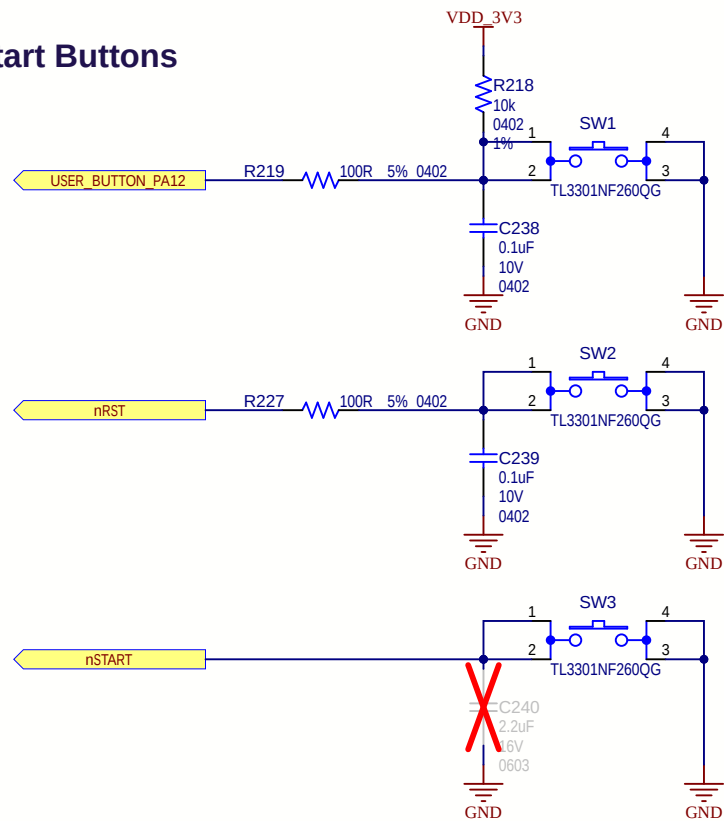


Drawn By: CAV			
Engineer: CAV			
PartNumber: EV21H18A	Project Title SAMA7G54 Evaluation Kit		
Sheet Title Audio Interfaces	Variant Name Standard	Designed with 	
Size B	SCH #: 03-00038	Rev: 5	Date: 3/1/2022 4:28:21 PM
	PCB #: 04-11152	Rev: 5	Sheet 13 of 15
File: 12 - Audio Interfaces.SchDoc			

RGB LED

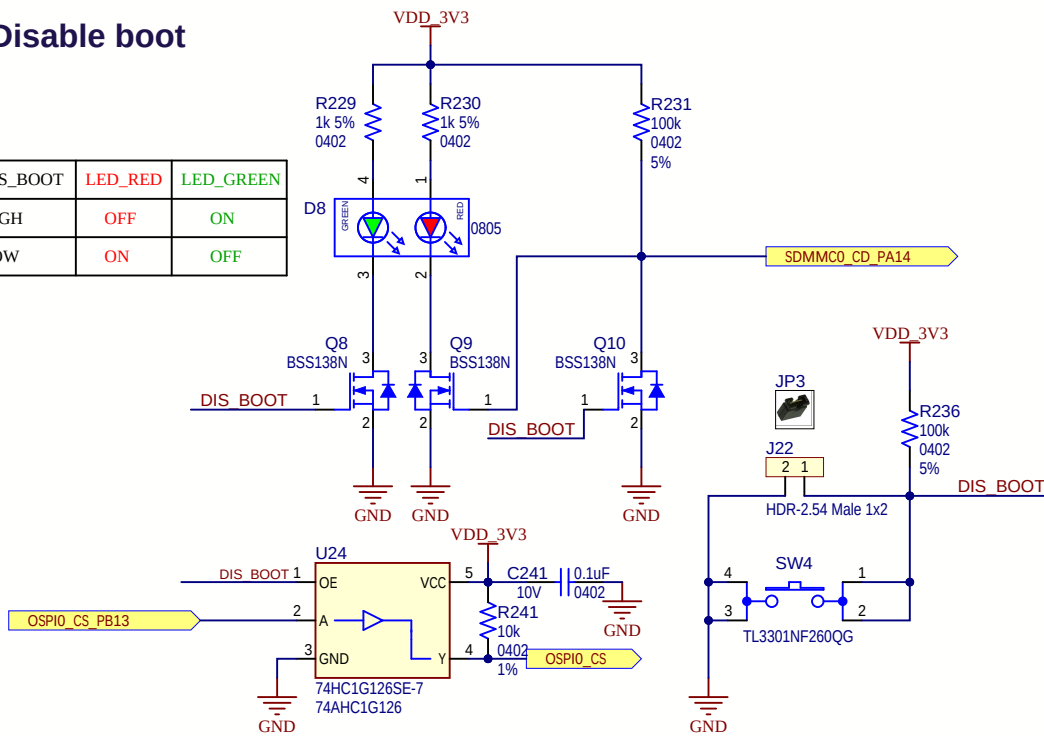


User, Reset, Start Buttons

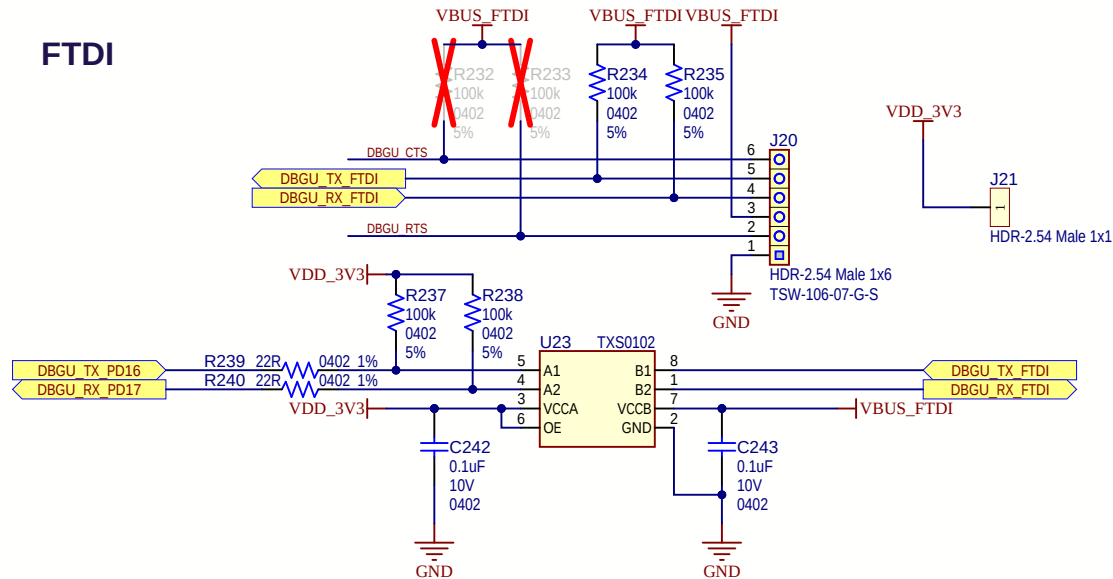


Disable boot

DIS_BOOT	LED_RED	LED_GREEN
HIGH	OFF	ON
LOW	ON	OFF



FTDI



Accessories



RUBBER PAD D9.5 H4.8



RUBBER PAD D9.5 H4.8



RUBBER PAD D9.5 H4.8



RUBBER PAD D9.5 H4.8



USB A to USB Micro-B Cable



USB A to USB Micro-B Cable



Need Help? Label

Drawn By:
CAV

Engineer:
CAV

PartNumber:
EV21H18A

Sheet Title
User Interface

Size B

File: 13 - User Interface.SchDoc



Project Title
SAMA7G54 Evaluation Kit

Variant Name
Standard

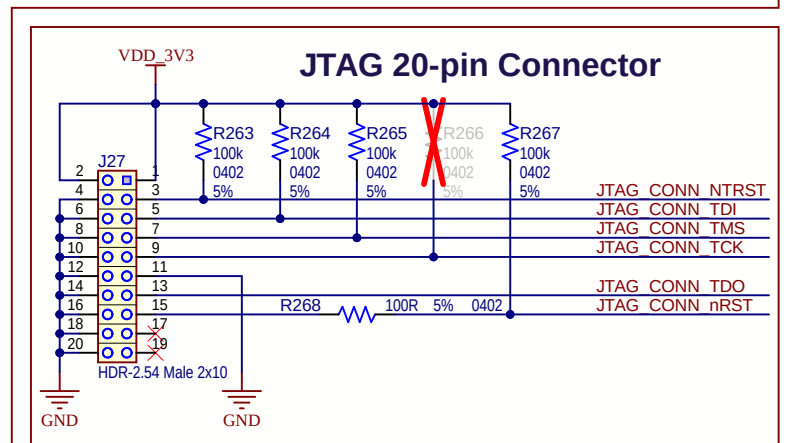
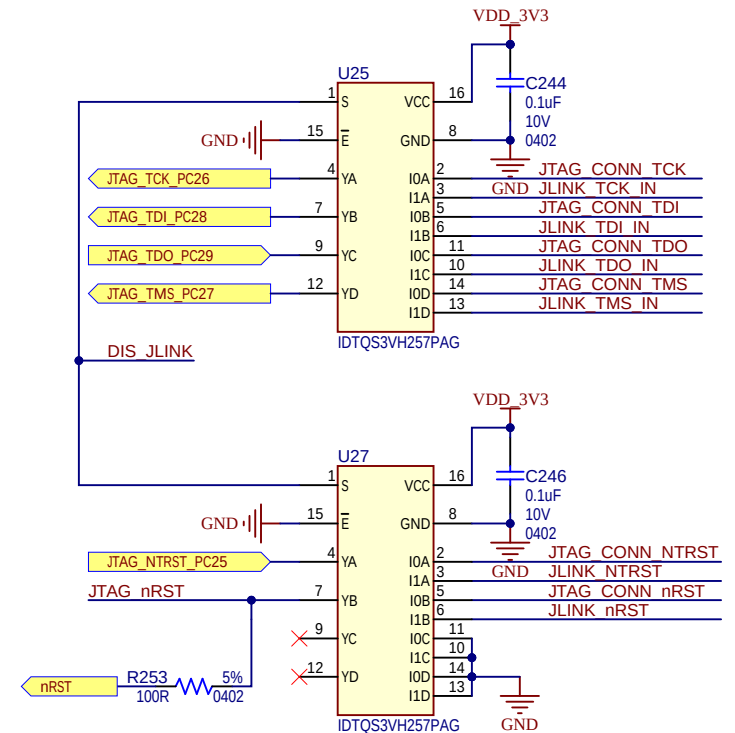
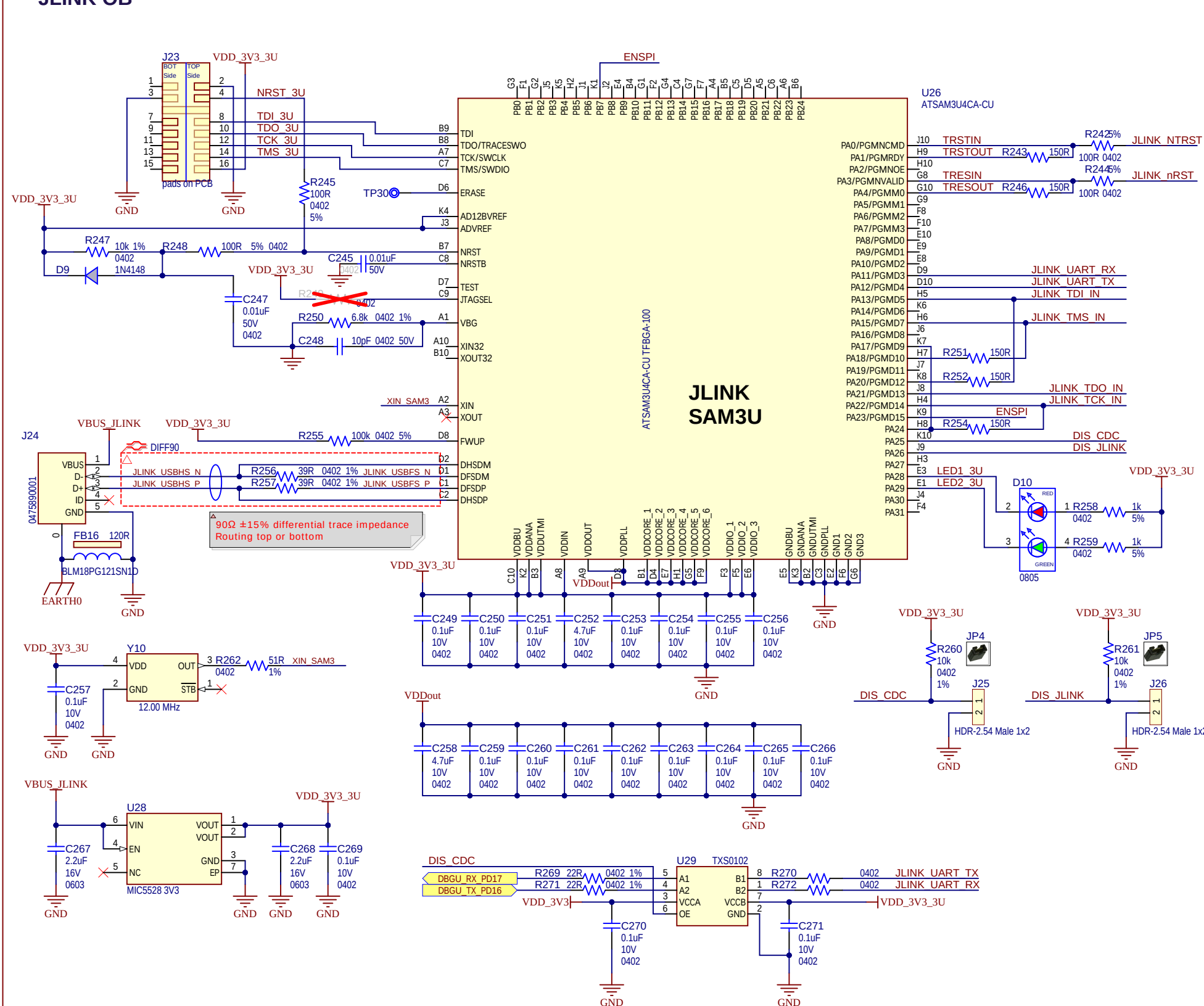
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Sheet 14 of 15

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JLINK OB

Drawn By: CAV		 MICROCHIP	
Engineer: CAV			
PartNumber: EV21H18A		Project Title SAMA7G54 Evaluation Kit	
Sheet Title JLINK Debugger		Variant Name Standard	
Size B	SCH #: 03-00038	Rev: 5	Date: 3/1/2022 4:28:21 PM
	PCB #: 04-11152	Rev: 5	Sheet 15 of 15
File: 14 - JLINK Debugger.SchDoc			
		Designed with  Altium.com	